

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

TRAINING TR-102 REPORT DAY 10

6 August 2025

Overview:

Today's session was packed with some core topics in AI and web development. I learned about how machines can recognize patterns and make decisions using algorithms like K-Nearest Neighbors (KNN) and Convolutional Neural Networks (CNN). We also learned about LAMP, a widely used stack in web development. Each of these plays an important role in building intelligent systems or dynamic websites, and the day gave us a nice balance of AI and backend development concepts.

Learning Objectives:

- Understand how **KNN** works and how to use it for classification.
- Learn how **CNNs** process images and extract features for AI applications.
- Explore the **LAMP stack** and its components in web development.
- Discover real-life applications of both **AI algorithms** and **server setups**.

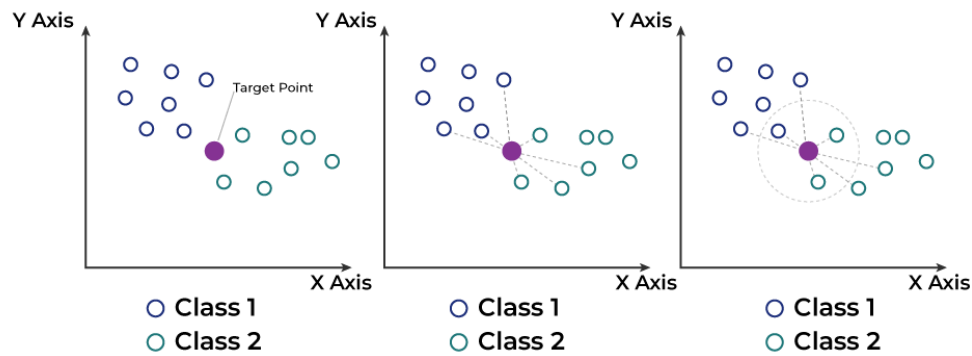
K-Nearest Neighbor(KNN)

KNN is a supervised machine learning algorithm used mostly for classification problems (and sometimes for regression).

It works by comparing a new input with the existing data points and selecting the 'k' closest neighbors (usually based on distance like Euclidean).

Working of KNN algorithm:

The K-Nearest Neighbors (KNN) algorithm operates on the principle of similarity where it predicts the label or value of a new data point by considering the labels or values of its K nearest neighbors in the training dataset.



Step 1: Selecting the optimal value of K

- K represents the number of nearest neighbors that needs to be considered while making prediction.

Step 2: Calculating distance

- To measure the similarity between target and training data points Euclidean distance is used. Distance is calculated between data points in the dataset and target point.

Step 3: Finding Nearest Neighbors

- The k data points with the smallest distances to the target point are nearest neighbors.

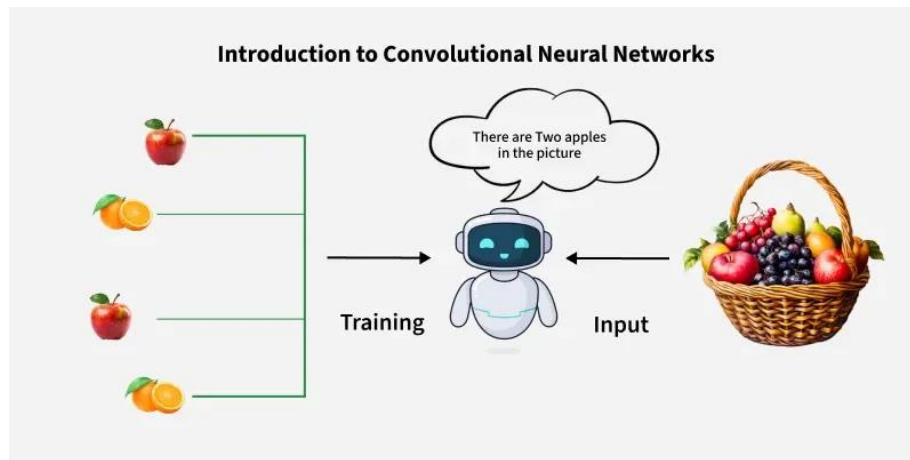
Step 4: Voting for Classification or Taking Average for Regression

- The algorithm still looks for the K closest points. But instead of voting for a class in classification, it takes the average of the values of those K neighbors. This average is the predicted value for the new point for the algorithm.

Example: Predicting if some fruit is an apple or orange based on size and color.

Convolutional Neural Network (CNN)

Convolutional Neural Networks (CNNs) are deep learning models designed to process data with a grid-like topology such as images. They are the foundation for most modern computer vision applications to detect features within visual data.



Working of CNN:

1. **Input Image:** CNN receives an input image which is preprocessed to ensure uniformity in size and format.
2. **Convolutional Layers:** Filters are applied to the input image to extract features like edges, textures and shapes.
3. **Pooling Layers:** The feature maps generated by the convolutional layers are down sampled to reduce dimensionality.
4. **Fully Connected Layers:** The down sampled feature maps are passed through fully connected layers to produce the final output, such as a classification label.
5. **Output:** The CNN outputs a prediction, such as the class of the image.

Example: Face detection, handwriting recognition, medical image analysis.

LAMP Stack

The LAMP stack is a popular open-source web development platform. It's a collection of software components that work together to build and manage websites and web applications. The LAMP stands for Linux, Apache, MySQL, and PHP.

Components:

- **Linux** The operating system that serves as the foundation for the entire stack.
- **Apache:** A web server that handles requests from users' web browsers and delivers web content.
- **MySQL:** A database management system that stores and manages the data for the web application.
- **PHP:** A scripting language used to create dynamic web pages and interact with the database.

Working:

1. A user's browser sends a request to the web server (Apache).
2. Apache forwards the request to PHP, which processes the request, potentially interacting with the database (MySQL).
3. PHP generates the dynamic content and sends it back to Apache.
4. Apache delivers the content to the user's browser.

Install LAMP Stack in Linux

Step 1: Checking Linux Distribution:

```
command : grep -E '^(VERSION|NAME)= ' /etc/os-release
```

Step 2: Upgrading Linux distribution to the latest packages.

```
command : sudo apt update && sudo apt upgrade
```

Step 3: Verify Apache Web Server Installation

Command: `apache2 -v`

Step4: Installing PHP

command : `sudo apt install php -y`

Conclusion:

Today's session was an eye-opener as we moved deeper into both machine learning and web backend. KNN taught us how machines make predictions by simply comparing with neighbors. CNN was exciting and impressive—how a machine can recognize faces or objects just by learning patterns in images. On the other hand, LAMP introduced us to the backbone of web servers, giving a real idea of how websites work from behind the scenes. A perfect blend of AI and development, this day gave us something useful in both fields.